

4.6 Astrophysics and cosmology

This topic covers the physical interpretation of astronomical observations, the formation and evolution of stars, and the history and future of the universe.

Students will be assessed on their ability to:		Suggested experiments
126	use the expression $F = Gm_1m_2/r^2$	
127	derive and use the expression $g = -Gm/r^2$ for the gravitational field due to a point mass	
128	recall similarities and differences between electric and gravitational fields	
129	recognise and use the expression relating flux, luminosity and distance $F = L/4\pi d^2$	application to standard candles
130	explain how distances can be determined using trigonometric parallax and by measurements on radiation flux received from objects of known luminosity (standard candles)	
131	recognise and use a simple Hertzsprung-Russell diagram to relate luminosity and temperature. Use this diagram to explain the life cycle of stars	
132	recognise and use the expression $L = \sigma T^4 \times \text{surface area}$, (for a sphere $L = 4\pi r^2 \sigma T^4$) (Stefan-Boltzmann law) for black body radiators	
133	recognise and use the expression: $\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m K}$ (Wien's law) for black body radiators	
134	recognise and use the expressions $z = \Delta\lambda/\lambda \approx \Delta f/f \approx v/c$ for a source of electromagnetic radiation moving relative to an observer and $v = H_0 d$ for objects at cosmological distances	
135	be aware of the controversy over the age and ultimate fate of the universe associated with the value of the Hubble Constant and the possible existence of dark matter	

Students will be assessed on their ability to:	Suggested experiments
136 explain the concept of nuclear binding energy, and recognise and use the expression $\Delta E = c^2\Delta m$ and use the non SI atomic mass unit (u) in calculations of nuclear mass (including mass deficit) and energy	
137 describe the processes of nuclear fusion and fission	
138 explain the mechanism of nuclear fusion and the need for high densities of matter and high temperatures to bring it about and maintain it	

Context-led approach (based on the Salters Horners Advanced Physics project)

The following section shows how the specification may be taught
using a context-led approach.